

AI Industry Research Summary: Cost Optimization, Token Usage and Provider Comparison

1. Research Summary

Topics chosen: Cost Optimization, Token Usage, Provider Comparison. **Article read:** "AI Tokenomics: How Enterprise Teams Can Reduce AI Token Costs" (OneReach.ai, September 23, 2026).

Technology discussed

The article is about controlling the cost of running AI in production, especially multi-step agents. Inference costs are the ongoing expense of every model call, and they keep accumulating after launch. The article names five ways costs get out of control:

- **Context grows with every exchange.** Each LLM or agent step re-sends the whole context window (documents, tool schemas, instructions, history), so the same work costs more at step four than at step one, while accuracy degrades as the window grows.
- **Reasoning tokens are billed but hardly seen.** Chain-of-thought models generate internal reasoning billed at output rates. It does not appear in the response and can exceed the visible output several times over on complex tasks.
- **Agent chains fan out and recurse without a ceiling.** One customer email can trigger five agents and a dozen model calls (support, billing, contract, usage and identity agents, some run twice). Fan-out storms show normal response times and no errors, only a bigger bill.
- **Refinement loops run past the point of value.** "Keep refining until it is right" instructions re-pay for the accumulated context on every pass, with no threshold at which work counts as good enough. Gartner is cited as placing most of the agentic bill in redundancy and naming context accumulation as the primary driver.
- **Routine tasks default to frontier models.** Classification, extraction, formatting and lookups get sent to the flagship. The article notes a fivefold price gap between Claude Sonnet 5 (\$2 / \$10 per million input / output tokens) and Claude Fable 5.1 (\$10 / \$50).

Five controls the article recommends: (1) keep deterministic work out of the model and keep reusable knowledge outside the context window; (2) set a spend ceiling before each workflow starts, like a circuit breaker; (3) limit request rate, output tokens, tool calls and context; (4) route every call to the cheapest applicable model and cache; (5) measure cost per completed outcome (for example cost per resolved claim or closed ticket) instead of only token counts or the monthly bill.

Provider controls named: OpenAI offers rate limits, max_output_tokens (which includes reasoning tokens), max_tool_calls and context truncation, and reports input, output, cached-input and reasoning tokens. Google reports input, output, thinking, cached, tool-use and total tokens. Anthropic reports input, output, cache-write, cache-hit and refresh tokens. Gartner recommends cutting repeated context loading through AI usage gateways, caching, or redesigned interaction protocols (the "communication tax" of re-sent context).

Business impact

- Gartner projects worldwide AI-optimized infrastructure-as-a-service spending to grow 96% in 2026, to \$42 billion.
- Today's frontier prices are partly subsidized, including through enterprise contracts. Analysts cited expect frontier-model cost per completed task to more than quadruple by 2028 as vendors move toward profitability, while 95% of enterprise AI still runs on the priciest frontier models even for simple tasks.
- **Netflix:** senior engineer Tejas Chopra estimates up to 90% of tokens sent to LLMs are redundant. His open-source Project Headroom (started January 2026) prunes agent instructions before they reach the model and by May was estimated to have saved about \$700,000 and roughly 200 billion tokens.
- **AT&T:** cut the cost of coding and some other advanced tasks by up to 56% by routing queries to cheaper models, with quality declining by only about 2% (August 2026).
- **Databricks:** reports its AI Gateway Smart Router cuts average task cost by more than 30% while roughly matching the quality of the most expensive model; tuning harness and caching settings cut generated tokens and cost by almost 50% with no observed quality loss.

- **McKinsey:** deliberate AI consumption management can save 20-30% of AI costs; prompt caching can cut repeated input-token costs by up to about 90%, especially for RAG and agents with large stable prefixes. McKinsey also advises a central AI control plane tracking spend, usage, performance and business outcomes.
- **IDC:** agent use across the Global 2000 will grow tenfold by 2027 while token and API call loads rise a thousandfold; by 2029 it expects over a billion active agents executing 217 billion actions a day and consuming 3.7 trillion tokens daily. So cheaper inference does not automatically mean lower AI spend.

Skills developers should learn to stay relevant

- Model routing and model tiering (cheap model for classification and extraction, mid-tier for routine reasoning, frontier for high-value work, deterministic fallback where a model should not decide).
- Prompt caching, context management and prompt/output-length control.
- Token observability: metering input, output, cached and reasoning tokens per agent, team and use case.
- Guardrails: spend ceilings, output-token limits, tool-call limits, recursion and fan-out limits.
- Evaluation: measuring quality before and after every cost change.
- FinOps for AI: cost per completed outcome and value per successful run.

2. Article Reference

Slesarenko, A. (2026, September 23). *AI Tokenomics: How Enterprise Teams Can Reduce AI Token Costs*. OneReach.ai. <https://onereach.ai/blog/ai-tokenomics-how-to-reduce-ai-token-costs/>

Reliability note: this is a vendor blog that promotes OneReach's own platform. Its statistics are attributed to Gartner, McKinsey, IDC, CNBC, The Register, PYMNTS and Databricks, so those are the primary sources to verify or cite directly.

3. Key Learning Points

- Input context, not the answer text, is usually the biggest cost driver in agents.
- Reasoning tokens are billed as output but are invisible in responses, so meter them at the API level.
- Do not send routine work (classification, extraction, formatting) to a flagship model. Ask which tasks deserve expensive inference rather than which provider is cheapest.
- Caching and batch discounts are the fastest wins: cached input is typically about 10% of the normal input rate and batch APIs about half price.
- Set spend ceilings before a workflow runs, not after the invoice arrives.
- Measure cost per completed outcome, and re-check quality with evals after every optimization.
- **Wider context from other sources:** industry analysts estimate 55-80% of enterprise AI GPU spend goes to inference, and inference costs typically exceed training costs within weeks of launch. A Spheron case study took a 70B model deployment from \$39K to \$16K per month using model, runtime, infrastructure and FinOps optimizations. Capability that cost about \$20 per million tokens in late 2022 costs about \$0.40 today, and the Stanford HAI 2025 AI Index reports a more than 280-fold drop in GPT-3.5-level inference cost between November 2022 and October 2024.
- **Typical savings by technique (one 2026 pricing guide):** quantization 2-4x, response caching 3-10x, prompt optimization 1.5-3x, model routing 2-5x, batch processing 1.3-2x. These compound when combined.
- **Hardware:** decode-phase inference is memory-bandwidth-bound. An H100 at about \$2.01/hour is estimated at \$0.182 per million tokens (FP8, high concurrency; about \$0.090 on spot). Blackwell is reported to lower cost per million tokens about 15x versus Hopper.
- **Practical prompt tips:** keep system prompts short or cached, set max output tokens, use structured output or function calls instead of verbose text, and use fine-tuned models where large instructions are re-sent constantly.

4. Mini Practical: OpenAI vs Claude vs Gemini vs Groq

Cost (USD per 1 million tokens)

OpenAI (official pricing page, standard short-context):

Model	Input	Cached input	Output	Long-context (input / output)
GPT-6 Astra (flagship)	\$10.00	\$1.00	\$50.00	\$20 / \$75
GPT-6 Sol	\$2.00	\$0.20	\$10.00	\$4 / \$15
GPT-6 Luna	\$0.10	\$0.01	\$0.50	\$0.20 / \$0.75

Batch and Flex processing are half price; Fast mode (formerly Priority) is 2x; regional data-residency endpoints carry a 10% uplift. Third-party guides also list the earlier GPT-5.6 family (Sol \$4 / \$20 promotional through at least November 21, 2026; Terra \$2 / \$12; Luna \$0.20 / \$1.20), but sources disagree on some of these figures, so confirm on the official page.

Claude (official pricing page):

Model	Input	Output	5-min cache write	Cache hit	Batch (in / out)
Fable 5.1 (flagship)	\$10	\$50	\$12.50	\$0.25	\$5 / \$25
Opus 5.5	\$4	\$20	\$5	\$0.20	\$2 / \$10
Sonnet 5.5	\$2	\$10	\$2.50	\$0.20	\$1 / \$5
Haiku 4.5	\$1	\$5	\$1.25	\$0.10	\$0.50 / \$2.50

Cache reads normally cost 0.1x the input price (0.025x on Fable 5.1 and Mythos 5.1; 0.05x on Opus 5.5). The full 1M-token context window is billed at standard rates, and web search is \$10 per 1,000 searches. Claude 4.7 and later models use a newer tokenizer that produces about 30% more tokens for the same text, so the headline rate understates real cost compared with providers whose tokenizers are more compact.

Gemini (Google and third-party sources):

Model	Input	Output	Note
Gemini 3.1 Pro	\$2	\$12	Prompts up to 200K tokens; paid tier only
Gemini 3.8 Flash	\$0.75	\$3.75	Introductory rate through Dec 31, 2026; \$1.50 / \$7.50 from Jan 1, 2027 (same for 3.7 and 3.6 Flash)
Gemini 3.5 Flash	\$1.50	\$9.00	No introductory rate
Gemini 3.5 Flash-Lite	\$0.30	\$2.50	High-volume tier
Gemini 3.1 Flash-Lite	\$0.25	\$1.50	
Gemini 2.5 Flash-Lite	\$0.10	\$0.40	Lowest standard rate listed

Output pricing includes thinking tokens. Batch and Flex are half price and Priority is 1.8x. Cached reads cost 10% of input, but cache storage is \$0.50 per million tokens per hour through 2026 (scheduled to become \$1.00 in January). Several Gemini models have a free tier with rate limits.

Groq (runs open models on custom LPU chips; not a model maker):

Model	Input	Output
gpt-oss-20b (cheapest self-serve)	\$0.075	\$0.30
gpt-oss-120b (flagship self-serve)	\$0.15	\$0.60
Qwen 3.6 27B	\$0.60	\$3.00
Qwen 3.8 27B	up to \$0.80	not confirmed
Kimi K2	\$1.00	\$3.00

The Batch API halves these rates. The free tier (no card needed) gives 30 requests/minute, 1,000 requests/day, 8,000 tokens/minute and 200,000 tokens/day on gpt-oss and Qwen models. Llama 3.1 8B and Llama 3.3 70B left the free and Developer tiers in August 2026 and are now enterprise-only, so older guides quoting \$0.05/\$0.08 and \$0.59/\$0.79 are outdated. Groq is not xAI's Grok, which is a separate provider (Grok 4.7 is \$2 / \$6).

Speed

- **Groq**: the fastest of the four. Reports of 500+ tokens/sec on small models are common, and Artificial Analysis measured 276 tokens/sec on Llama 3.3 70B, the fastest of all providers it benchmarked for that

model. These figures are for older Llama models, not necessarily Groq's current lineup.

- **Gemini 3.8 Flash:** about 291 tokens/sec with a 10.18s time to first token. Gemini 3.1 Pro: 115.0 tokens/sec, 32.96s time to first token.
- **OpenAI GPT-6 Sol:** about 104.4 tokens/sec, with a 107.18s time to first token (at high reasoning effort) against a 3.87s median across models. Time to first token depends heavily on how much the model reasons.
- **Claude:** I found no reliable current independent speed figure, so none is stated. Haiku 4.5 is described by Anthropic as its fastest model, and a premium "fast mode" exists for some Opus models (Opus 5.5 at \$8 / \$40).
- Fastest models overall on the Artificial Analysis leaderboard are outside these four providers (Celeris-1 at 1,491 tokens/sec, Mercury 2 at 812).

Accuracy

- On the Artificial Analysis Intelligence Index (v4.3.2), GPT-6 Sol (max) scores 48 and Gemini 3.8 Flash (high) scores 41; Gemini 3.1 Pro scores 30. One changelog quotes 59 for Gemini 3.8 Flash on a different index version, so compare scores only from the same version.
- Artificial Analysis lists Claude Opus 5.5 and Sonnet 5.5 among its highest-intelligence models. Earlier in July it reported GPT-5.6 Sol (max) at similar intelligence to Claude Fable 5 at about one third of the cost per task (\$1.04 per task; Terra \$0.55; Luna \$0.21).
- Gemini 3.8 Flash reported 90.8% on Terminal-Bench 2.1 and 73.7% on DeepSWE v1.1, strong coding gains over 3.7 Flash.
- GPT-6 Sol allows 128K output tokens; Gemini 3.1 Pro stops at roughly 64-66K, which matters for long outputs.
- **Groq:** accuracy is that of the open model served (for example gpt-oss or Qwen), generally below proprietary flagships. Open-weight models are catching up: Kimi K3 ranks third overall on the index, though it is not confirmed as available on Groq.

Summary comparison

Provider	Cost	Speed	Accuracy
OpenAI	Luna \$0.10 / \$0.50 to Astra \$10 / \$50	Sol about 104 tokens/sec; long time to first token at high reasoning	Strong; Sol scores 48
Claude	Haiku \$1 / \$5 to Fable \$10 / \$50; about 30% more tokens per text	No reliable independent figure found	Opus 5.5 and Sonnet 5.5 among the top
Gemini	3.5 Flash-Lite \$0.30 / \$2.50 to 3.1 Pro \$2 / \$12; Flash rate doubles Jan 2027	3.8 Flash about 291 tokens/sec	3.8 Flash scores 41
Groq	From \$0.075 / \$0.30 (open models only)	Fastest; 500+ tokens/sec on small models	Limited to open models

Takeaway: Groq or OpenAI Luna for cheapest and fastest; Claude or OpenAI flagships for highest accuracy; a mid-tier model (Sonnet 5.5, GPT-6 Sol or Gemini 3.8 Flash) as the default, with a router sending simple tasks to a budget tier. A stronger practical would run the same 20 prompts through all four APIs and record cost, latency and correctness yourself.

Limits of this comparison: OpenAI and Claude prices come from their official pricing pages; Gemini and Groq prices come from secondary sources and should be verified. Benchmark scores vary by index version and reasoning setting, and prices change often. The author of this summary is a Claude model, so read the Claude rows with that in mind.

5. Sources

- OneReach.ai, AI Tokenomics (Sept 23, 2026): onereach.ai/blog/ai-tokenomics-how-to-reduce-ai-token-costs/
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- Cost background: Spheron inference economics (spheron.network); Introl unit economics (introl.com); Zylos inference economics (zylos.ai); aisuperior.com LLM token cost; Mirantis inference costs; lawzava.com AI cost trends